A Text-as-Data Approach to Measuring Interstate Relations Using National Security Strategies*

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How states regard one another – which states command attention, and with what evaluative orientation – is central to international relations, yet it is measured with proxies that are largely binary, symmetric, and slow-moving because they are built for other purposes. I develop and validate a text-as-data method that recovers directed dyad-year indices of a state's expressed salience (prominence) and affinity (evaluative stance) toward other states from its national security strategy. The method automates the judgments a careful reader would make – which states a government names, how prominently, and in what evaluative terms – and makes them consistently across hundreds of documents. Applied to a new public corpus of 438 national security strategy documents issued by 92 countries (1962–2024) – to date the most comprehensive cross-national, multi-decade collection of security-strategy texts – the approach produces continuous, directional, and time-varying

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measures of interstate posture. Because governments revise these documents every few years rather than every few decades, the measures register shifts in stated posture as they happen, an early signal alongside slower-moving indicators such as UN votes and alliances. The measures converge with benchmarks such as alliance portfolio similarity and UN voting patterns where they should, track human judgments of expressed orientation, and diverge from the benchmarks where expressed and behavioral alignment part ways. In application, the scores add incremental predictive validity for militarized interstate dispute onset and diplomatic visits beyond these benchmarks, and reveal an asymmetry – cooperation tracks both a target's prominence and its warmth, while conflict tracks prominence most strongly, with expressed hostility contributing a smaller association. The corpus, processing code, and derived measures are publicly released, providing a replicable tool for extracting conveyed relationships from political texts.

1 Introduction

In its 2017 National Security Strategy (NSS), the United States described Taiwan in a single passage as part of the section on regional-security. By comparison, the 2022 edition named Taiwan seven times across two chapters, including its treatment of China, declaring "an abiding interest in maintaining peace and stability across the Taiwan Strait" and committing to "maintain our capacity to resist any resort to force or coercion against Taiwan" (2022, 24) – language widely read as a firmer deterrence commitment (Mastro 2023). This time period experienced no observed change in the relationship in terms of treaty, alliance commitment, or formal recognition. What moved is how much attention Taiwan commanded in each NSS and in what register.

Nor does the signal run only from Washington. In its 2013 White Book, Ukraine's Ministry of Defence listed among its priorities "developing a strategic partnership with the Russian Federation, the United States of America and the People's Republic of China on the basis of effective and mutually beneficial cooperation" (2013, 51). Within a year, Russia had annexed Crimea. Russia does not recede from the White Books that follow – it remains the most prominent state in nearly every subsequent edition – but the terms change: partnership language migrates toward NATO and Western capitals, and mentions of Russia now invoke a phrase never mentioned prior to 2014: "the military aggression of the Russian Federation" (2015, 4). Attention held constant but the expressed orientation had inverted.

The standard measurement toolkit has no place to put either movement. Taiwan has cast no UN vote since 1971 and joined no alliance since 1980, so instruments that infer affinity from voting agreement or alliance portfolios record nothing there. And no widely used indicator separates how much attention a state pays a target from how it regards that target – much less whose regard it is. Yet both movements were legible in official public security documents at the

moment they happened. This paper is interested in understanding the information states use to convey their relationships toward other states. These signals are systematically recoverable from states' own strategic documents in a way that the standard measures of interstate ties are not built to carry.

What other states does a state talk about and how does it orient itself toward each of them? I call the first dimension *salience* – how much of a state's security attention another state commands – and the second *affinity* – the evaluative orientation, from hostility to warmth, that it expresses toward that state.[1] These two dimensions are foundational to international relations theory: they shape who counts as politically relevant (Maoz 1996; Bennett 2006), which dyads cooperate and which fight (Leeds et al. 2002; Diehl, Goertz, and Gallegos 2021), and how scholars draw inferences about foreign policy alignment (Gartzke 1998; Signorino and Ritter 1999; Bailey, Strezhnev, and Voeten 2017; Voeten 2026). Yet empirical work has had to rely on proxies that are coarse and indirect. Political relevance is typically a binary indicator built from geography and great-power status that varies little within a dyad over time. Affinity is typically inferred from alliance portfolios, UN voting patterns, or rivalry codings, but these instruments are designed for other purposes that often treat the relationship as symmetric and slow-moving. Each captures something real, but none captures how a state itself articulates the prominence of, and orientation toward, another. After all, states are not generally turning to scholarly quantifications of UN votes or alliance similarities to infer changes in interstate relationships. Existing approaches to signaling in international relations have largely focused on intentions, capability, and resolve (Fearon 1994; Fuhrmann and Sechser 2014; Gartzke et al. 2017; Min 2025), leaving the relational content of these texts – who a state conveys is important, and how positively or negatively it speaks of them – largely outside the measurement enterprise.

[1] Throughout, *expressed*, *conveyed*, and *articulated* are used interchangeably to describe the salience and affinity a state communicates about other states in its national security strategy.

That articulation lives in states' own texts and can be recovered systematically. This paper develops and validates a text-as-data method to produce a directed dyad-year index of two distinct constructs: the salience a state assigns to another (how prominent the target figures in the originating state's text) and the affinity it expresses toward that target (the evaluative stance, from hostile to friendly). The estimates are produced by an automated pipeline (Figure 2) that finds every passage in which a strategy names another state, classifies the stance that passage expresses toward it with a fine-tuned natural language inference model, and grades the intensity of that stance with a transparent sentiment lexicon. Each step is validated. The stance classifier reaches a Matthews correlation of 0.925 on held-out hand-coded excerpts; the resulting affinity scores correlate with out-of-sample human rankings of evaluative tone at $r = 0.68$ ($n = 100$); and affinity scores survive a full translation round trip (English to Spanish, French, or Russian and back) at $r = 0.90$–0.91, matching the $r = 0.91$ at which the pipeline reproduces its own published values on untranslated text. The pipeline is applied to a new corpus of 438 National Security Strategy (NSS) documents issued by 92 countries (1962–2024) – strategic documents that are intended in part for foreign audiences, crafted with care over which states to name, how often, and in what terms. These are also demanding documents for such an exercise: vetted, bureaucratically produced, and often dismissed as cheap talk (Lissner 2017), so relational signal recovered from them under explicit validation standards is a hard test of the approach rather than an easy one.

The paper makes three contributions. Methodologically, the pipeline combines off-the-shelf natural language processing (NLP) and natural language inference (NLI) components in a way that generalizes beyond this corpus: other scholars can apply it to diplomatic cables, parliamentary speeches, party manifestos, or any other corpus where the analyst wants to recover who is being talked about and how. It joins a broader effort to extract otherwise-latent political quantities from text (eg. Grimmer and Stewart 2013; Fariss et al. 2015; Lauderdale and Herzog 2016; Baturo, Dasandi, and Mikhaylov 2017; Grimmer, Roberts, and Stewart 2022; Carson,

Min, and Nuys 2024), with the specific contribution of recovering directed, target-aware relational content rather than topic or sentiment alone. Empirically, the NSS corpus, processing code, and derived measures are made publicly available at (https://militarydoctrines.com/) as a community resource, to date the most comprehensive cross-national, multi-decade collection of public security-strategy texts, building on and extending prior compilations (Becker and Malesky 2017; Razeto and Jenne 2021; Becker et al. 2025; Neal and Gardner 2026). Substantively, the resulting measures are consistent with existing benchmarks while providing distinct information. On a common measure of directed dyad-years, salience and affinity have incremental predictive validity for both Militarized Interstate Dispute (MID) onset and diplomatic visits even when accounting for commonly used measures like alliance-portfolio similarity, UN voting similarity, and the most commonly used measures of politically relevant dyads. These measures are also asymmetric: both salience and affinity matter for cooperative engagements (diplomatic visits), while conflictual engagements (MID) track prominence most strongly, with expressed hostility contributing a smaller association. This provides systematic evidence that states' own articulations of the salience and affinity of other actors carry consequential information that scholar-imposed proxies and measures of other material state actions do not. For policy analysis, this makes the measures a candidate early signal: they register shifts in a government's stated posture on the publication cycle of its strategy documents, often before those shifts appear in slower-moving indicators such as UN votes and alliance portfolios.

The methodological premise deserves stating plainly. Machine coding political texts is now ubiquitous and will be for some time; what remains scarce is awareness of the source material worth coding. Governments routinely produce standardized, comparable, public statements of how they see their security environment, and the more one knows about how governments work and what they publish, the more measurement opportunity that knowledge creates. The contribution here is accordingly not any individual classifier, as better components will keep arriving, and the released pipeline is built so they can be swapped in. Instead, it is the pairing

of a deliberately assembled corpus with a validation architecture that makes the recovered quantities credible.

The remainder of this paper proceeds as follows. Section 2 reviews existing approaches to measuring salience and affinity in international relations and where they fall short. Section 3 sets out the two-dimensional conception of interstate ties that the measure operationalizes, including descriptions of the NSS corpus, its assembly, and its public release. Section 4 sets out the modeling pipeline that produces the directed dyad-year measures. Section 5 validates the estimates against canonical benchmarks of political relevance and foreign-policy similarity. Section 6 applies the measure to predicting militarized interstate disputes and diplomatic visits as a test of predictive validity. Section 7 concludes with limitations of the corpus and approach, and directions for extending both.

2 Existing Measurements of Interstate Ties

Initially motivated by a concern about which interstate dyads constituted the most appropriate universe of empirical cases, much ink has been spilled both theorizing and measuring what makes a state relevant to the security of another (Maoz 1996). The prevailing view takes a gravity model approach, assuming the opportunity and willingness of two states to undertake meaningful political engagement is a function of their geographic distance and material power/status.[2] Early work thought about geographic distance as some form of direct or indirect contiguity (Weede 1976; Bennett 2006), minimum distance (Gleditsch and Ward

[2]My aim is not to rehash debates about 'politically relevant dyads' here, nor to provide another reformulation of what dyads share a propensity for conflictual or cooperative political interaction. For detailed evaluations of various criteria for 'political relevance', see in particular Lemke and Reed (2001), Benson (2005), Bennett (2006), Xiang (2010), and Braumoeller and Carson (2011). For extensions beyond the 'political relevance' framework, more tailored understandings include dangerous dyads (Bremer 1992), politically active dyads (Quackenbush 2006, 2024), strategic reference groups (Maoz 2010), globally relevant dyads (Lee and Thompson 2017), extra-dyadic linkages (Gartzke and Gleditsch 2022), and grievance dyads (Owsiak and Vasquez 2022). For the turn toward network-based models of political relevance see, among others, Maoz (2010), Ward, Ahlquist, and Rozenas (2013), Dorff and Ward (2013), and Cranmer and Desmarais (2016).

2001), capital-to-capital distance (Lemke 1995), or co-location in the same region (Small and Singer 1982). Measures of material power and status similarly abound, with many arguing a dyad is politically relevant if it contains a great power (Organski and Kugler 1980; Singer 1988; Maoz and Russett 1993) and others using more granular measures of force projection like the Composite Index of National Capabilities (CINC) (Singer, Bremer, and Stuckey 1972) or military spending (Kydd 2000).

Broadly speaking, existing research has considered affinity as potential or actualized threats or partnerships. Despite being a bipolar scale, much work justifiably focuses on either the positive or negative dimension of affinity to hone theoretical richness.[3] Some scholarship has highlighted historical instances of hostility and rivalries (Thompson 2001; Terechshenko 2020; Thompson, Sakuwa, and Suhas 2021), while others identify and theorize formal alliances and similar forms of defense cooperation (Leeds et al. 2002; Gibler 2009; Kinne 2020). Given the unavoidable entanglement between positive and negative interstate relations, others have developed ways to simultaneously measure the full spectrum of affinity using dyadic peace levels ranging from serious rivalry (most negative) to security community (most positive) (Diehl, Goertz, and Gallegos 2021). The most common measures of international affinity conceptualize it as foreign policy similarity measured using alliance portfolios (Signorino and Ritter 1999; Chiba, Johnson, and Leeds 2015) or UN voting patterns (Gartzke 1998; Bailey, Strezhnev, and Voeten 2017), with recent work inferring latent state preferences from networks of cooperative and conflictual ties jointly (Gallop and Minhas 2021; Beardsley 2024; Edgerton 2024).

Looking more broadly, these literatures measure three distinct concepts: *preferences*, what a state wants, usually inferred from revealed behavior such as votes; *relationships*, standing ties such as alliances and rivalries; and *behaviors*, what states actually do to one another. The measures developed in this paper fit none of these categories exactly. They capture the

[3]This section is by no means exhaustive since, as is implied by its name, the entire field of international relations is about variation in international relationships.

expressed relationship: the prominence and evaluative orientation a government chooses to convey about another state when it publicly describes its security environment. An expressed relationship is related to underlying preferences and to observed behavior but reducible to neither – a state may voice warmth it does not feel, or leave unstated a commitment it fully intends to honor – and the gaps between what is expressed, what is wanted, and what is done are informative in their own right. Treating expression as its own object of measurement is what allows the divergences documented below to be read as signal rather than error.

These measures are widespread for good reason; the data have been carefully collected, they often have broad geographic and temporal scope, and findings using these data are consistent with much of what scholars intuitively suspect about international interactions. But while empirically tractable, these approaches tend to assume symmetry, relative temporal stability, and/or are often binary or ordinal measures that make more nuanced distinctions between observations challenging.[4] In reality, some allies may be indispensable while others are peripheral; some rivals dominate security planning while others are ignored. Given the binary nature of measures of political relevance, it is primarily used as a methodological scoping condition rather than something that calibrates the intensity of expressed affinities. Furthermore, states themselves do not produce these datasets and it is unlikely that government decisions are influenced by how interstate relations are coded by scholars. So, indirect scholarly measures of how states relate to each other capture something similar, but not identical, to how states communicate their relations to each other. Establishing that a measure recovered from those communications improves on these benchmarks, rather than merely differing from them, requires holding it to explicit standards of measurement validity (Adcock and Collier 2001): convergence with established indicators where they should agree, and informative divergence

[4]All these shortcomings do not apply to every measure. UN voting patterns do exhibit temporal variation, but have been criticized for masking strategic abstentions (Bailey, Strezhnev, and Voeten 2017), inflating consensus (Häge and Hug 2016), and statistical difficulties accounting for chance agreement (Häge 2011). Alliances have well established observable differences in their depth, reliability, and reliance (Mattes 2012; Alley 2021; Gannon 2026). Rivalries are similarly distinguished as being "lesser" or "severe" (Diehl, Goertz, and Gallegos 2021) and in terms of their cause (Thompson, Sakuwa, and Suhas 2021).

where it captures what they cannot. The validation in Section 5 proceeds on those terms.

3 A Two-Dimensional Schema of Expressed Interstate Ties

How a state regards another decomposes into two analytically distinct dimensions. The first is salience: how much of the issuing state's attention the target commands, identified by its prominence in the state's articulation of its security environment (cf. Posner 2004 on salience as the contextually variable political importance of an actor or cleavage). The second is affinity: the evaluative orientation the issuing state expresses toward the target, from hostility to warmth. The two are conceptually independent. A target can be highly salient and regarded negatively (an existentially-driven rival), highly salient and positively regarded (an indispensable ally), or peripheral and regarded with either notable affinity or disinclination, or, most likely, indifference conveyed as neutrality. Existing measures rarely account for both dimensions simultaneously: political-relevance indicators capture a coarse, largely binary version of salience while saying nothing about affinity, and affinity proxies such as alliance portfolios or UN-voting similarity capture valence irrespective of whether states themselves know or care about the voting patterns and alliance similarities of other states. When both are included, salience is typically relegated to a scope condition of binary political relevance, and what a specific affinity measure tells us about geopolitical alignment is underspecified (Voeten 2026). There is a parallel with recent research in the US domestic context that similarly decomposes a familiar composite construct (affective polarization) into distinct subdimensions (othering, aversion, and moralization) and finds those subdimensions have distinct relationships with key phenomenon that are lost when using the composite measure (Campos and Federico 2025).

3.1 Overview of the Data and Analyses

The empirical work draws on two bodies of data. The first is the corpus itself: national security strategies and equivalent whole-of-government documents, collected, processed into machine-readable text, and publicly released; the analytic subset comprises the English-language, whole-of-government documents described in the next section. The second is a set of gold-standard human judgments: independent coders ranked text excerpts by expressed evaluative orientation using best–worst scaling, providing the human benchmark against which the machine-coded measure is validated and the labeled anchor for bias-corrected estimation of the relationships it enters.

3.2 NSS Corpus

I define an NSS as a public text written by a state outlining aspects of their foreign policy pertaining to security and defense. These documents matter; they guide and describe national security decisions by connecting ends and means and by conveying intended national security priorities (Caudle 2009; Ettinger 2017; Johnstone 2024). Policymakers invest significant resources in their production and attend closely to those issued by others, in part because these documents exist to be appraised and scrutinized by foreign audiences (Snider and Nagl 2001; Mohr et al. 2013; DuMont 2019). These texts are intended to provide relational insight, functioning "as strategies for relating enemies and allies" (Burke 1941, 304). Indeed, because threat assessment itself is "at its heart, a form of storytelling," NSS documents serve as one of the primary vehicles for such narratives (Krebs 2015, 16). These characteristics explain why, "if any words matter, the words in [national security strategy documents] do" (Chin, Skinner, and Yoo 2023, 104). The corpus can thus be thought of as a sample of public documents in which states communicate the importance of other international actors and their orientation toward them.

The first stage involved compiling 438 NSS documents (1962–2024) from 92 states via online government archives, think tank repositories, and multilingual secondary databases, building on efforts by Becker and Malesky (2017), Razeto and Jenne (2021), Becker et al. (2025), and Neal and Gardner (2026).[5] Metadata was also collected to inventory the document name, issuing country, year of publication, and language of publication. In cases where the document was a specific subcategory — such as a cybersecurity strategy or budget outlay — this was also recorded. Figure 1 shows the temporal and linguistic distribution of the corpus-eligible documents. More descriptive information about the spatial distribution, per-country publication counts, and the overlap with the closest comparable corpora are in the Data Coverage section of the Appendix.

[5]The most comparable alternate corpora are the Neal & Gardner National Security and Defence Documents Dataset v3.5 (Neal and Gardner 2026), which covers 671 documents from 118 countries (1987–2025) and Becker et al. (2025). The full militarydoctrines.com release reaches further back in time and covers a partially overlapping set of states; a country-year comparison of the two corpora is in the appendix.

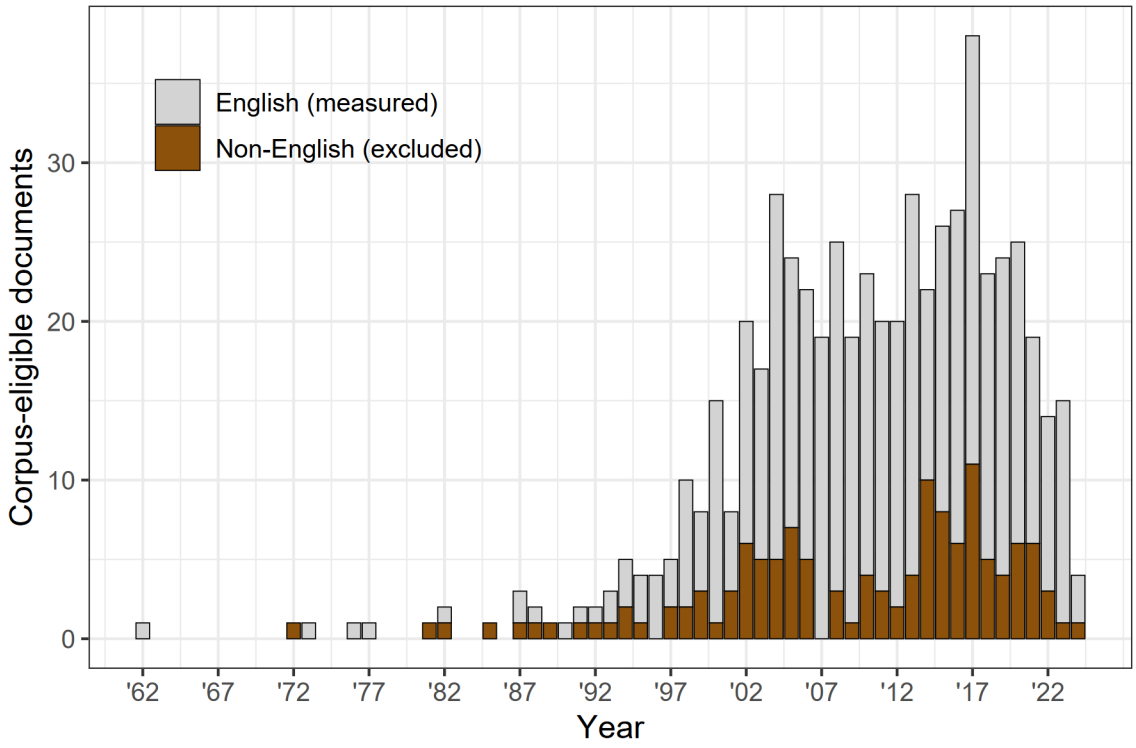


Figure 1: Corpus coverage by year of publication and language. Light bars are the English-language documents from which the measures are built; dark segments are the non-English documents the language filter excludes.

In the second stage, I convert the pdf files to markdown and plaintext format using PyMuPDF4LLM (Smith 2023; McKie 2024) and Marker (Paruchuri 2024). This conversion process preserves formatting markers demarcating natural linguistic boundaries like headings, paragraph breaks, lists, and the location of images and tables to maintain context provided by the structure and order of the documents. The csv version of the data is thus formatted with each page of the pdf as a distinct row and columns providing the markdown text as well as identifier information about the country, year, document name, and page number. This then allows each NSS to be separated into textual segments (chunks) using an MPNet tokenizer from 'sentence-transformers/all-mpnet-base-v2' (Song et al. 2020).[6]

[6]The alternatives to chunking often pre-process texts, which fails to retain contextual information (Lahoti,

The result is a dataset of NSS documents in pdf, plain text, markdown, and csv format that can be viewed and downloaded in bulk or individually at (https://militarydoctrines.com/). The code used to convert the documents is also provided there so that future improvements in text processing, particularly in non-English languages and of non-Latin characters, can use the existing corpus as a benchmark for measuring improvements. I anticipate future work by other scholars can examine the issue-specific documents, compare security strategies issued by different agencies within a government, and study the degree to which non-English language or non-Latin character documents systematically differ in how they are processed. But as a note of caution, the comparability of these documents across space and time must be theoretically informed by knowledge of the document's specific context and the research question being explored (Stolberg 2012). Not all states produce these documents, and those that do often do so in some years, but not others. This could occur because a state produces these documents at regular temporal intervals like once per administration, because of strategic reasons to keep such information secret, or because a state has insufficient bureaucratic capacity to produce them (or even insufficient concern for international security). Alternatively, states may produce more than one type of NSS and may even do so in the same year.[7]

To ensure comparability across texts, I focus on the NSS for the U.S. as "the only complete whole of government national security document that the U.S. Government publishes" (Stolberg 2012, xi) and make similar choices for other states when more than one document exists for a given year. While acknowledging the significant diversity in state capacity, political systems, and bureaucratic traditions that shape these documents, this study proceeds on the premise that when states choose to publicly issue a formal 'National Security Strategy' or equivalent high-level defense White Paper, they are engaging in a broadly similar *type* of

Garimella, and Gionis 2018; Burnham 2024a; Overbeck et al. 2025) and has recognized biases when used with the more advanced computational models used here (Palomino et al. 2017; Denny and Spirling 2018; Miyajiwala et al. 2022).

[7]For example, the United States publishes a National Security Strategy (NSS) as its primary document, but also publishes a National Defense Strategy (NDS), National Military Strategy (NMS), Nuclear Posture Review (NPR), National Homeland Security Strategy (NHSS), and National Intelligence Strategy (NIS).

strategic communication aimed, at least in part, toward international audiences. The core function of articulating security priorities and framing key relationships is assumed to be a common denominator, though the specific nuances and emphases will vary.

The closest-analogue rule also admits genre variation. For most states the analogue to the U.S. NSS is a strategy statement; for some, the most comprehensive public security document is a national threat assessment or an intelligence service's annual report, and several states publish annual defense reviews rather than episodic strategies. These genres carry their own rhetorical conventions, and the conventions move the measures: a threat assessment names even close partners inside threat sentences, so Lithuania's 2021 security strategy scores Germany at $+0.97$ while its 2023 national threat assessment scores the same ally at -0.94. The corpus metadata records each document's title and type, so users can condition on genre; comparisons that pool across genres inherit this heterogeneity.

Consequently, the indices of salience and affinity that constitute the remainder of this paper are calculated using only English-language NSS documents that represent a 'whole-of-government' security strategy most analogous to the U.S. NSS; issue-specific national security documents like those concerning only cyber defense or budget outlays are excluded. The majority of NSS documents are published in English, as it often serves as a lingua franca for international diplomatic communication and states issuing an English version of their NSS likely intend it for a broad international audience. And methodologically, while multilingual LLMs are advancing (Waight et al. 2025; Griswold, Robbins, and Pollard 2026), achieving high fidelity across languages using the computational models used here presents significant challenges, and ensuring comparable measurement quality across languages for the indices of interest is a substantial undertaking beyond the scope of this study.

The size and scope of this footprint can be stated precisely. Of the 584 corpus-eligible documents, 130 (about 22%) are published exclusively in a non-English language. Restricting the

sample to only English-language documents removes 18 states from the corpus entirely — a group dominated by 13 Latin American countries — and thins the document count for 35 more, Russia's Russian-only strategies among them. The retained corpus therefore over-represents states that address international audiences in English, and cross-national comparisons in what follows should be read as statements about that selected population rather than about all strategy-publishing states. Advances in multilingual stance detection suggest the pipeline could in future be extended to native-language strategies directly (Mets et al. 2024), although NLI models to measure stance and a weighting mechanism for cross-lingual sentiment are still needed.

I test whether the non-English documents structurally differ from the English language documents in a manner that affects measurement. Five states in the corpus publish the same strategy document in both a native language and an official English translation (France 2013, Germany 2016, Russia 2009, Chile 2002, and Finland 2021). I machine-translate each native original with open-source OPUS-MT neural translation models and run both the official-English and translated-English versions through an identical measurement pipeline (Tiedemann and Thottingal 2020). Across the 161 target countries the paired documents mention, dyad-level affinity computed from the machine translation tracks affinity computed from the official English at $r = 0.83$; among the 54 targets mentioned at least three times in both versions, $r = 0.87$ with a mean absolute difference of 0.14 on the $[-1, 1]$ scale. As a benchmark for those magnitudes, running the same replica pipeline on the official English text and comparing against the published corpus values yields $r = 0.93$. I conclude from this that translating non-English language documents before running them through the pipeline adds roughly as much measurement error as incidental implementation details of the pipeline do. The English-only restriction therefore operates chiefly as a limit on coverage rather than a distortion of what is measured, and machine translation offers a practical route to extending the corpus to the excluded documents.

3.3 The Data-Generating Process

The corpus just described is not a random sample of interstate communication, but the product of intentional government decisions that together tell prospective users of the data exactly which population each measure describes. The first decision is whether to publish at all. States may publish no strategy due to the structure of government, low bureaucratic capacity, interest in secrecy, or absence of a salient international security environment. The second decision concerns the language. Knowing any publication can be translated, the decision to issue an official English edition is a choice that conveys the intended audience. From this, I infer that the measures capture posture as expressed *to international audiences* for whom these documents are, at least in part, produced. Third, the interval. Publication is episodic with state-specific rhythms. Some states publish once per administration, others once every half decade, others at moments of security reorientation, and others with no discernible pattern. The result is an unbalanced panel whose gaps vary within as well as across countries (documented in the Data Coverage section of the appendix). The release does not interpolate between publication events, because a strategy speaks for the moment of its issue; whether and how to fill inter-publication years is an analytical choice that belongs to the user and the research question, not to the dataset. Last is whom to name. Within a published document, naming is itself a choice. Silence about a state is informative about attention (its absence) but affinity is undefined without evaluative text. Consequently, unnamed targets carry no affinity score.

Together, these four decisions define the population the measures describe: states that choose to address international audiences in an English-language, whole-of-government strategy, in the years they choose to publish, concerning the states they choose to name. That population is narrower than the universe of dyad-years covered by universal-membership benchmarks like UN voting, but it is also precisely the population for which the constructs are defined.

Salience and affinity are properties of expressed posture, and expression requires a document, a language, a moment, and a named target.

4 Part I: Measuring Salience and Affinity

For each state in the Correlates of War Project (2017) state membership list, I create a list of aliases and synonyms that include variations of the country name, capital city, and demonyms.[8] I then use these lists to identify the chunks in the NSS documents where at least one non-self state is mentioned. The chunks are then pruned into sentences using spaCy's sentence tokenizer to identify sentences where State B is mentioned as well as the sentence immediately before and after that reference (Montani et al. 2023). This process allows me to identify all mentions of each state in the NSS documents and their immediate semantic context.

Applying text-based machine learning techniques to this processed data, I produce a continuous directed dyad-year measure of the two dimensions introduced in Section 3 – salience and affinity – from each issuing state toward each target state referenced in the issuing state's NSS (n = 14,145). Salience captures the target's prominence in the issuing state's NSS; affinity captures the evaluative orientation the issuing state expresses toward it. This procedure is summarized in Figure 2 and explained in the sections that follow.

[8]States may reference each other indirectly (e.g. 'our northern neighbor') or as collectives (e.g. 'our European allies'). Given the number of texts, possible references, and ambiguity about appropriate inclusion involved in properly coding such references, I here only include direct country-to-country references. Future work using more advanced NLP techniques and area-specific knowledge could provide additional coding for country-region and country-institution references like NATO and the European Union as well as study the conditions under which states use intentionally ambiguous references like euphemisms or amalgamations (Jee et al. 2024).

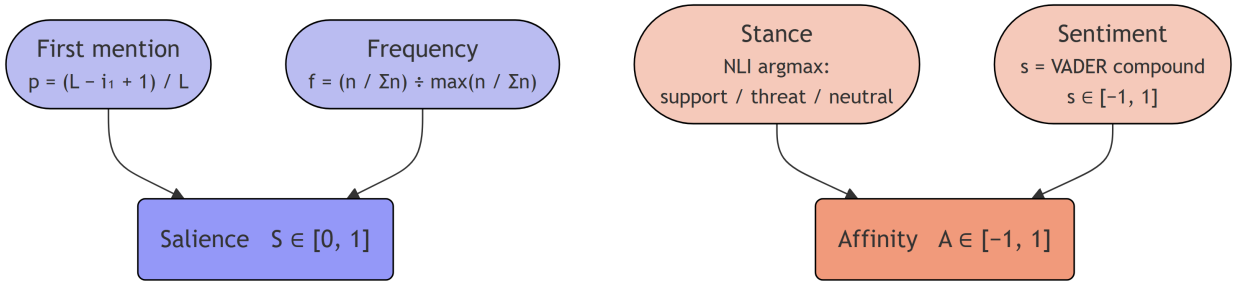


Figure 2: Components of salience and affinity. Salience sums an equally weighted first-mention position and mention-frequency share onto [0, 1]; affinity combines a fine-tuned NLI stance label with VADER sentiment intensity through a banded fusion onto [-1, 1].

4.1 Model and Assumptions

The measurement model treats each NSS as the observable product of an underlying directed posture. When a government drafts its strategy, it holds some latent regard toward each other state — how much that state occupies its security planning, and with what orientation — and that regard shapes concrete drafting choices like how prominently to name a state (if at all) and in what evaluative language to discuss it. The pipeline inverts this process, recovering the two latent dimensions from the drafting choices they generate: two observables for prominence (first mention and mention frequency) identify salience, and two observables for evaluative language (target-directed stance and sentiment) identify affinity.

Three assumptions connect the observables to the constructs. First, NSS text is deliberate, internationally-audienced communication. The drafting choices are informative because they are made intentionally and reviewed at the highest levels, as the documents' production process and reception suggest (see Section 3.2). Second, the salience observables are monotone in latent salience: states are not systematically obtuse in referencing actors they consider most central, an assumption consistent with the documents' explicitly communicative function. Third, stance-weighted sentiment is monotone in latent affinity: warmer expressed language

about a target reflects a warmer conveyed orientation. None of these assumptions requires one to equate expressed posture with private belief; the constructs are defined as *expressed* salience and affinity, and the validation in Part II tests them as such expressions.

The scales for each variable are conventionally anchored. Salience is normalized to $[0, 1]$ within document, combining a first-mention percentile with a frequency share, so it is a relative measure of attention allocation within each strategy. Affinity is anchored directionally by the stance classification — the NLI step fixes which direction is "positive" — and scaled to $[-1, 1]$ by the banded fusion with sentiment. As baseline prose tone could differ systematically across countries or time, a cross-country anchoring adjustment in the appendix re-centers each origin's scores against its own pooled distribution, and the applications include flexible temporal controls so that inferences do not ride on scale drift across eras.[9]

4.2 Salience

First, I measure the *salience* of a target entity in the issuing state's NSS text chunks using two indices that are commonly used in computational models of search engine optimization (SEO) and salient named entity extraction (SNEE) both independently and in combination; first-mention and entity-frequency (Dunietz and Gillick 2014; Wu et al. 2020; Grönberg 2021).[10]

First-mention is the document chunk index m_e for each target entity e where it is first mentioned. I then identify the total number of chunks N where at least one entity is mentioned and invert and normalize the measure as $S_{\text{first}} = \frac{N - m_e + 1}{N}$ so that entities mentioned in the first chunk have a score of 1 and entities mentioned in the last chunk $m_e = N$ have a score

[9] The cross-country anchoring adjustment is modeled off of work re-anchoring legislative effectiveness scores and interest group ratings (Groseclose, Levitt, and Snyder Jr. 1999; Anderson and Habel 2009; Volden and Wiseman 2014).

[10] Other salience detection models like SWAT (Ponza, Ferragina, and Piccinno 2019), SEL (Trani et al. 2018), and transformer models (Asgarieh, Thadani, and O'Hare 2024; Bullough et al. 2024) can achieved higher precision and recall, but produce only binary outputs and are very sensitive to fine-tuning, impacting replicability and external validity (Bhowmik et al. 2024).

strictly greater than, but approaching zero.[11] This normalization adjusts for the fact that NSS documents vary greatly in length and some states mention other actors frequently while others mention other states infrequently.[12]

Entity-frequency is measured as the total number of chunks f_e where entity e is mentioned. This is calculated as $S_{\text{freq}}(e) = \frac{f_e}{f_{\max}}$ where $f_{\max}$ is the highest number of chunks any single entity appears in within that same document. This normalization similarly adjusts for the fact that some issuing states mention many states in their NSS and others mention relatively few. Because the data are processed in pruned chunks rather than individual words or sentences, multiple references to a country within the same contextual chunk do not bias the measure.

First-mention and entity-frequency are both bounded $x \in (0, 1]$ with higher values indicating higher salience. Salience is then measured by linearly combining both components as $S_{\text{salience}} = \alpha S_{\text{first}} + (1-\alpha) S_{\text{freq}}$ where $\alpha = 0.5$ is a hyper-parameter equally weighting the two components and maintaining the original scale. This reflects the theoretical assumption that both initial prioritization and sustained attention contribute equally to a state's communicated strategic focus within these documents.[13] Whereas political relevance is typically operationalized by scholars as a binary property, salience here represents a continuous, directed measure of stated importance from one state toward another.

[11] An alternate measure using $S_{\text{first}} = \frac{N - m_e}{N}$ would mean that states first mentioned in the last chunk of a text would have a first-mentioned score of 0. The use of $m_e + 1$ allows me to differentiate these states from those that are not mentioned at all, as the latter are more appropriately measured as having a salience of 0 than the former. The notation that follows assumes that non-mentioned entities are ignored and then assigned a score of 0 if it is a recognized COW state in that year.

[12] This approach implicitly assumes that the relative position within a document (e.g., appearing halfway through) carries similar weight regardless of absolute document length, effectively normalizing for reader attention across documents of varying sizes. While well-resourced state apparatuses might scrutinize entire documents meticulously, the attention of broader audiences or analysts in less critical dyads might indeed wane with document length. This assumption is a simplification, and alternative normalization schemes could be explored in future research.

[13] For robustness, I also calculate, and then similarly scale a multiplicative combination $S_{\text{salience}} = S_{\text{first}} \times S_{\text{freq}}$ which provides a higher penalty for low values on either dimension.

4.3 Affinity

Second, I measure *affinity*, the affective position expressed toward a target state. Although many computational text analysis models have used sentiment to measure attitude toward an entity, more recent work has demonstrated that sentiment conveys something sufficiently distinct from stance that the former cannot accurately proxy for the latter (Mohammad et al. 2016; Rauh 2018; Boukes et al. 2020; Chan et al. 2021; Küçük and Can 2021; van Atteveldt, van der Velden, and Boukes 2021; Bestvater and Monroe 2023; Burnham 2024b).

Unfortunately, current state-of-the-art stance metrics produce only binary measures, and measurement uncertainty cannot be easily aggregated across multiple text chunks mentioning the same target entity. Since recent methodological work identifies exactly this categorical limitation and recommends continuous stance measures, I calculate affinity using a combination of sentiment and stance (Griswold, Robbins, and Pollard 2026).

For *sentiment*, the evaluative language concerning a target state is processed using VADER, a well-established rule-based sentiment analyzer (Hutto and Gilbert 2014).[14] In this context, 'sentiment' refers to the positive, negative, or neutral evaluative orientation of the language used, rather than an imputation of a collective emotional state to the entity issuing the document. VADER assigns an initial continuous score $x \in [-1, 1]$ for each text chunk mentioning a given target state.

Then, *stance* is measured using recent advances in few-shot learning and target-aware opinion

[14] As classifying the *direction* of an evaluation toward a named target is the more challenging enterprise, it is handled by a fine-tuned modern classifier (Section 4 below). The more tradition lexicon-based VADER is instead confined to grading the *intensity* of language within a stance band already assigned by the classifier. I choose this lexicon-based approach over a transformer model because the preference is for a transparent rule set that is deterministic and fully reproducible without reference to a hosted model checkpoint. This decision is also close to inconsequential since substituting a modern transformer sentiment model reproduces the affinity measure at $r = 0.84$ across all 157,175 corpus chunks and with no improvement in validation against the human labels (Section 4). The z-scored robustness check in the appendix correlates with the headline measure at $r \approx 0.95$.

classification to capture sentiment directed toward a distinct entity rather than the overall affect conveyed in a text chunk (Biber and Finegan 1988; Du Bois 2007; AlDayel and Magdy 2019). In the context of NSS documents, the issuing state conveys a stance toward a target state which may be distinct from the general sentiment being expressed in that particular document chunk. This form of stance detection is done using natural language inference (NLI), also known as textual entailment, which uses language models as pre-trained universal classifiers to identify the stance of a text toward a particular entity (Burnham 2024a).

Stance is detected using a fine-tuned classification model defining a directional relationship between a text T and a hypothesis H about whether a human reading T would infer that H is true (A. Wang et al. 2020; Küçük and Can 2020; AlDayel and Magdy 2021; Burnham et al. 2024; Laurer et al. 2024).[15,16] In this context, the model calculates a stance classification as a Boolean response to the proposed hypothesis H that the issuing state's stance toward the target state is positive, negative, or neutral. Given the Boolean output, the model is run separately for each hypothesis H (threat, neutral, or support) and the classification with the

[15]Further technical details are provided in the appendix but in brief, I employ a few-shot classification approach by fine-tuning an LLM (Political DEBATE, based on DeBERTa V3, which outperforms BERT- and RoBERTa-based classifiers on political-science text-classification tasks (Timoneda and Vera 2025)) on a small set of hand-coded document excerpts from the corpus (Burnham et al. 2024). These metrics are computed on an in-domain held-out set: hand-coded NSS excerpts from the same corpus that were not used to train the few-shot classifier. The deployed classifier whose configuration is used to label the full corpus achieves a Matthews Correlation Coefficient of 0.925 and an accuracy of 0.962 on that held-out set ($n \approx 26$). Because the classifier is few-shot, held-out performance depends on the training draw; the appendix reports its sensitivity across random seeds and over the full few-shot learning curve as an additional robustness check. For similar work using active learning to fine-tune models like RoBERTa to a specific local task, see Bonikowski, Luo, and Stuhler (2022) and Card et al. (2022).

[16]I fine-tune a small NLI classifier (Political DEBATE) rather than zero-shot prompting a frontier LLM because fine-tuning on hand-coded political text is inexpensive, deterministic, and fully replicable across users, and because the approach performs competitively on political stance-classification benchmarks (Burnham et al. 2024). Zero-shot prompting depends on a proprietary API whose outputs drift over time, and reliance on closed-source models raises transparency and replicability concerns for published research (Spirling 2023; A. Palmer, Smith, and Spirling 2024). A zero-shot comparison on a sample of chunks is a useful direction for future validation, but it is not the basis on which the present measure rests. Results on political texts favor the fine-tuned encoder: Maerz, Schafer, and Schneider (2026), measuring illiberal rhetoric across more than 400,000 paragraphs of leader speeches with fine-tuned DeBERTa models, cross-validate those scores against OpenAI's GPT-5.2 on a human-coded gold standard and report that the fine-tuned model matches or slightly exceeds the frontier model on both of their dimensions ($R^2 = 0.75$ versus 0.76, and 0.78 versus 0.71), with lower error and bias.

highest probability is selected. These results produce a stance score $x \in \{-1, 0, 1\}$, for each target state in each chunk.

A concrete example is helpful. Assume T states 'the death and destruction of innocent civilians in Ukraine is a horrific tragedy'. The language model is asked to classify the author's stance toward the target entity Ukraine. The first step proposes H: 'the author of this text expresses support or alliance with Ukraine.' Based on its reading of the text chunk, the model would classify the hypothesis as 'true' with a given probability. The process is repeated with the negative and neutral hypotheses and from those three, the one with the highest probability is selected. In this case, that would be the 'supportive' stance indicative of a positive affinity toward Ukraine irrespective of the sentiment of the words used in the text.

This appropriately captures the semantic relationship between the expression and target in a directional manner composed of an evaluated recognizable objective H, an expression T containing a positive, negative or neutral evaluation, and a true/false relation that attributes the expressed valuation to the object in question (Fogel-Dror et al. 2019; Overbeck et al. 2025). Put more plainly, what this does – and sentiment analysis alone does not – is identify an expressed stance toward a target entity even when the general sentiment in that text is incongruent with the sentiment toward the specific target entity or when the general sentiment provides no information about the author's stance toward the target entity.

The sentiment and stance detection values are then combined to produce the baseline measure of directed-dyad affinity. First, I align the valence of each chunk's sentiment score with its predicted stance: support/alliance predictions become positive, threat/rival predictions become negative, and neutral predictions remain unchanged. This adjustment accounts for cases like the Ukraine example where emotionally charged words appear in contexts contrary to their typical connotation. I then scale the scores into three distinct ranges: negative stances are $x \in [-1, -0.5]$, neutral stances are $x \in (-0.5, 0.5)$, and positive stances are $x \in [0.5, 1]$. This

scaling preserves the categorical stance information while allowing for differences in intensity within each category. The values are then averaged across all chunks for a given target state, producing a continuous value $x \in [-1, 1]$ reflecting both the direction and strength of the affinity.

The unit of evidence deserves a precise statement. Affinity is computed from the text immediately surrounding each mention of the target, so what the fused score measures is the stance expressed in the passages that name the target, aggregated over those passages. Usually that stance is the origin evaluating the target directly. Sometimes the target instead enters as the site of events, in a chronology, or inside the recounted position of a third party, and the evaluative language attaches to the target without asserting the origin's own orientation. The gold-standard coders whose labels anchor the validation in Section 5 read the same passages under the same instruction, so the measure and its validation share one scope: expressed stance in passages naming the target. Dyad-year affinity should be read with that scope in mind.

This banded fusion is robust to its two main design choices. First, the ± 0.5 band cutoffs are flexible: re-deriving the chunk-level affinity with cutoffs at ± 0.4 and ± 0.6 leaves its correlation with the human gold-standard labels essentially unchanged (Pearson r between 0.67 and 0.69 across the three settings). Second, although recent work has argued that lexicon-based approaches (LBA) like VADER are outperformed by recent large language models (LLMs), additional analysis replacing the VADER method with a transformer-based sentiment model yields a chunk-level affinity that correlates with the human labels just as well as the VADER-based measure ($r = 0.69$ versus 0.68, $n = 100$) and agrees closely with it ($r = 0.86$) (Griswold, Robbins, and Pollard 2026). That agreement is not an artifact of the small gold sample: re-scoring every one of the 157,175 chunks in the corpus with the transformer and re-fusing under the same scheme reproduces the VADER-based affinity at $r = 0.84$. The measure is therefore largely invariant to the choice of sentiment scorer, which is what the design below intends: the

scorer supplies intensity, and the fine-tuned classifier supplies direction. The validation of the fused chunk-level affinity against human judgment is reported in Section 5 and the appendix.

I benchmark these estimates of affinity against gold-standard human judgments elicited through Best-Worst Scaling (BWS), a survey method that produces reliable comparative rankings using human annotation (Finn and Louviere 1992; Louviere 2015; Kiritchenko and Mohammad 2016). Rating affinity on an ordinal scale has challenges of intercoder similarity and ties, but pairwise comparisons, though well validated for political-text coding (Carlson and Montgomery 2017), are computationally expensive in coder effort and the amount of codings needed. BWS takes the best of both of these and asks coders to choose rather than rate, a method that has been demonstrated to yield more reliable per-item scores from the same amount of coding effort (Kiritchenko and Mohammad 2017). In this task, human coders were presented with randomized tuples of four text chunks, each mentioning a specific target state, and identified the chunk expressing the most positive and the chunk expressing the least positive evaluative language toward that target.[17] Each chunk's gold-standard score is the share of its tuple appearances chosen as most positive minus the share chosen as least positive, pooled across coders; two coding rounds scored 100 chunks in total, with coder agreement reported in the appendix. These human scores are the held-out benchmark against which the fused affinity is validated (Section 5), and the labeled subset for design-based supervised learning (DSL), which corrects estimates of downstream relationships involving the machine-coded measure for its measurement error (Egami et al. 2024b, 2024a). Notably, the BWS coding is not used to rescale the affinity measure because doing so would make them a circular validation benchmark.

This validation-and-correction architecture is itself a methodological contribution. Design-

[17] Coders were given the target state to code for the particular entry (as some text chunks mention more than one state) and instructed to make their evaluation using only information in the chunk. By rating only the best and worst option from the set of four, 5 of the 6 pairwise comparisons are simultaneously coded with no additional coder effort. Further details on the BWS implementation and inter-coder reliability are provided in the Appendix.

based supervised learning is a recent development that combines a small set of gold-standard labels with the machine predictions available for every observation, yielding estimates of downstream relationships that are corrected for measurement error in the machine labels, with standard errors that remain valid despite the labels' imperfection. Principled bias correction of this kind is still rarely paired with text-as-data measurement in international relations, and it gives the affinity measure a defensible inferential anchor alongside the corpus itself and the decomposition of a state's posture into salience and affinity.

5 Part II: Validation

5.1 Face Validity

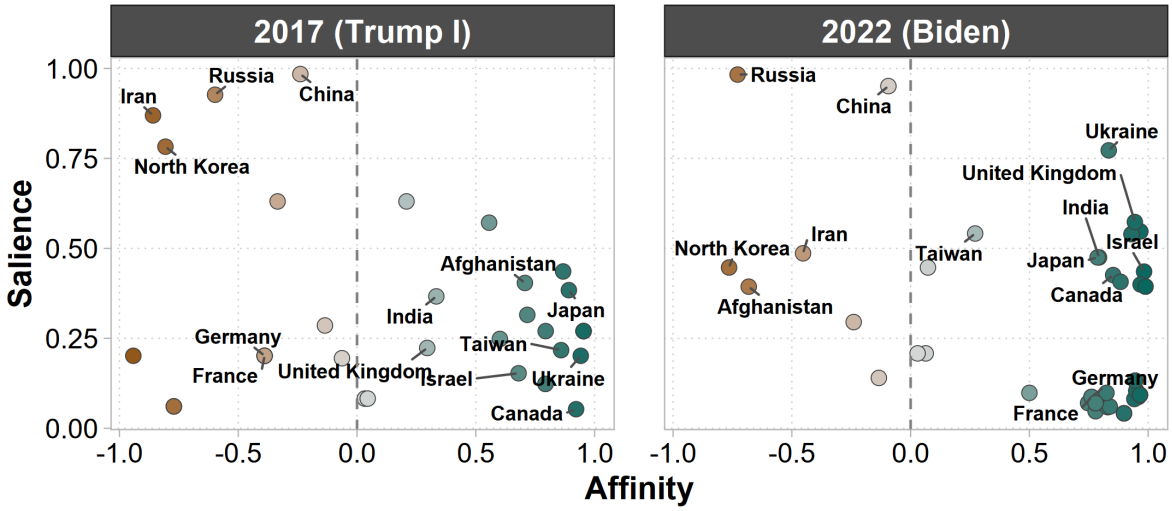


Figure 3: Affinity and salience for states named in the 2017 and 2022 US National Security Strategies. The same states are labelled in both panels to permit direct comparison. Because the United States is itself a major power, every state shown would conventionally be coded as a politically relevant dyad.

To illustrate the face validity of the measures of salience and affinity, Figure 3 shows the scores for states mentioned in the U.S. NSS issued during the first Trump administration (2017) and the Biden administration (2022). The data are consistent with the first Trump administration's "America First" strategy warning of great power competition with Russia and China as well as rogue states like Iran and North Korea, and his dissatisfaction with U.S. allies in Europe (Landler and Sanger 2017). This is contrasted with President Biden's 2022 NSS that identified US security priorities as long-term competition with China, an immediate threat from Russia, and a stronger emphasis on positive multilateral ties with traditional U.S. allies like the United Kingdom, India, and Japan (McDonald 2023). And although the affinity expressed toward Ukraine is positive in both documents, Ukraine's salience in the 2022 NSS (0.77) is much higher than in the 2017 NSS (0.2), reflecting a shift in how the US communicated the importance of Ukraine in its national security discourse.

This figure also visually demonstrates that salience and affinity function as largely independent dimensions: states with high, medium, and low salience scores can be found across the spectrum of affinity values.[18] Notably, some states exhibit high salience despite having a relatively neutral affinity score (e.g., values clustering near zero on the affinity x-axis but high on the salience y-axis), suggesting they are considered important but the issuing state conveys neither strong alignment nor animosity.

The two dimensions also separate over time within a single relationship. Figure 4 traces four decades of US strategy documents writing about China. Salience is essentially constant: China's prominence never falls below 0.34 and averages 0.68 across all 20 documents, so a binary coding of political relevance sees no movement — China is simply always relevant, in 1982 as in 2022. Affinity is not constant at all. It runs from -0.65 in 1982 through positive territory in the engagement era, peaks at 0.88 in 1991, and falls back below zero in 2017, changing sign 4 times along the way.

[18]The distribution of salience and affinity scores across the entire population is provided in the Appendix.

No measure built from alliance portfolios or UN voting recovers that trajectory because those aggregates move slowly and symmetrically. Separating the two dimensions is also what makes it legible. A composite that multiplied prominence by warmth would report the same near-zero value for 1990 and 2022 as for a state the United States never mentions, even though the first two describe an important relationship treated with reserve and the third describes no relationship at all.

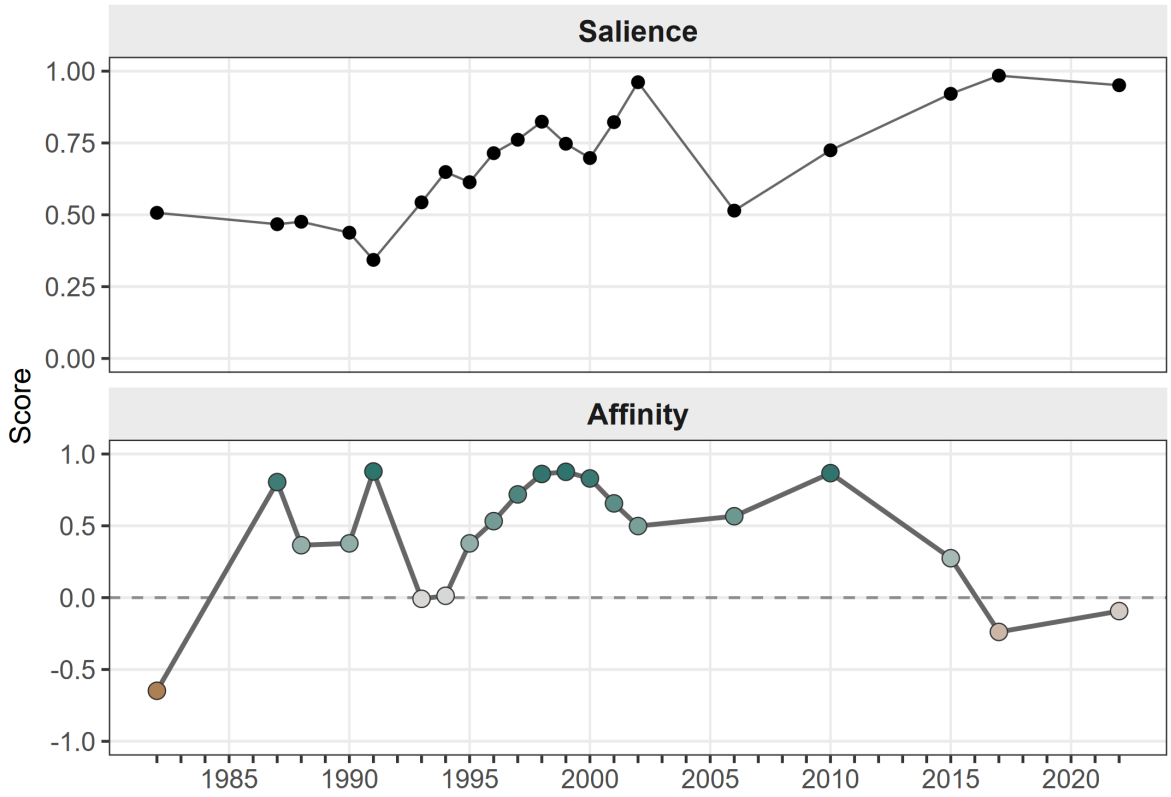


Figure 4: United States NSS scores for China, 1982–2022. Each point is one strategy document; the United States publishes irregularly, so the horizontal gaps are real and no values are interpolated.

Table 1 provides an illustrative example of both measures and their inputs for the 10 highest salience states in Ukraine's 2017 NSS. Not surprisingly, Russia is the most salient state in Ukraine's NSS, as it is the first country Ukraine references and it is also mentioned much

Table 1: Ten highest-salience targets in Ukraine's 2017 NSS, with the inputs behind each index. Salience is bounded [0, 1] and affinity [-1, 1]; higher values indicate greater salience and warmer expressed affinity.

	Salience			Affinity		
Target Country	First Mention	Entity Frequency	Salience	Sentiment	Stance	Affinity
Russia	1.00	1.00	1.00	-0.07	-0.89	-0.72
Poland	0.93	0.74	0.83	0.69	0.43	0.56
United Kingdom	0.98	0.58	0.78	0.70	0.36	0.53
Lithuania	0.93	0.42	0.67	0.65	0.38	0.51
United States	0.89	0.37	0.63	0.70	0.71	0.71
Germany	0.78	0.42	0.60	0.31	0.50	0.40
Norway	0.87	0.32	0.59	0.88	0.83	0.86
Denmark	0.96	0.11	0.53	0.66	0.00	0.33
Netherlands	0.98	0.05	0.52	0.76	1.00	0.88
Greece	0.76	0.26	0.51	0.57	0.00	0.29

more frequently than any other country. The U.S., by comparison, has a first-mention score of 0.89, and Ukraine mentions the U.S. roughly a third as often as it mentions Russia. Ukraine expresses negative affinity toward Russia and positive affinity toward the U.S. and other European states with notable variation across the positive cases.

5.2 Construct Validity

Second, I validate the measures by demonstrating their convergence with existing measures of salience and affinity, an established approach to validating text-derived measures (Bond and Messing 2015). Theoretical expectations posit that states should signal more positive affinity toward states with which they share common security interests, as arrangements like defense alliances are, in part, a means to "document, advertise or encourage such an affinity" (Gartzke and Weisiger 2013, 25). First, I identify the correlation between signals of affinity in NSS documents and the most well-regarded measures of foreign policy similarity: kappa-corrected

s-score measures of alliance portfolio similarity and UN voting patterns.[19] Conveyed affinity has a positive and statistically significant correlation with both measures (alliance portfolios = 0.14 and UN voting = 0.09).[20] For comparison, the correlation between alliance and UN voting kappa-corrected s-scores is 0.39. States publicly signal warmth toward others in patterns similar, but not identical to, what one would assume given similarity of defense partners and UN General Assembly voting patterns. Continuous ideal-point estimates from UN voting are an alternative benchmark that captures more nuance than kappa-corrected s-scores (Bailey, Strezhnev, and Voeten 2017); the same broad pattern of partial agreement holds against that comparison as well. In the framework of Adcock and Collier (2001), these are convergent/discriminant validation checks, and they are necessary but not decisive on their own: a correlation against imperfect proxies is inherently ambiguous, because a moderate association is equally consistent with an informative measure that diverges from flawed benchmarks and with a merely noisy one — and even a strong correlation with an established-but-imperfect measure would not settle the matter. I therefore read these correlations alongside a criterion validation against human judgment, presented next.

There is precedent for benchmarking text-derived political measures against human-coded labels (Lai et al. 2024; Griswold, Robbins, and Pollard 2026). Two rounds of independent coders ranked text chunks by expressed evaluative orientation using best–worst scaling, yielding 100 gold-standard human labels. Crucially, the affinity measure is constructed without reference to these labels: the human rankings are used only to validate the measure and, below, to bias-correct estimates of its relationships with the benchmarks; they never rescale the affinity values themselves. The correlation between the model's chunk-level affinity and the human gold-standard labels is therefore an out-of-sample validation, and it is high: Pearson $r = 0.68$

[19] I use the kappa chance-corrected versions of the s-scores (Cohen 1960; Häge 2011), but a full table of correlations using pi-corrected, binary, or weighted un-corrected measures as well as using COW rather than ATOP alliance data is provided in the appendix.

[20] These correlations are calculated for directed dyad-years where the salience of State B > 0. Dyads with no mention (and thus no NSS affinity score) are necessarily excluded from this specific comparison, as non-mention primarily speaks to a lack of expressed salience rather than a neutral affective tone.

(95% CI [0.56, 0.77], $n = 100$).

As a stricter check, restricting to the second round of labels held out of an alternative calibration fit on the first round alone also yields $r = 0.68$. A correlation of this magnitude against human coders, set beside the more modest correlations with alliance and UN-voting similarity, indicates that the affinity measure tracks human assessments of evaluative orientation closely while diverging from the scholarly aggregates in ways that reflect signal rather than measurement error. It is unlikely that attenuation explains the modest s-score correlations since correcting them for the NSS measure's estimation error (dividing by the square root of its held-out human-validation coefficient (≈ 0.68)) raises them only to about 0.16 (alliance) and 0.1 (UN voting). The limited convergence therefore reflects a genuine construct distinction rather than noise. The orientation a state *expresses* in its own strategy is related to, but not redundant with, the *revealed*, behavioral alignment encoded in alliance portfolios and UN General Assembly votes. That non-redundancy establishes the discriminant side of the validation alongside the convergent correlations above. Full details, including the scatter and a sensitivity analysis, are in the appendix.

These validations are, in substance, validations of the stance classification. The lexicon grades intensity only within a band the classifier has already assigned (Section 4), so the fused score moves with the stance labels, and a construction of affinity from the stance labels alone reproduces the application estimates of Section 6 (see the alternative constructions of affinity in the appendix). Claims about what affinity tracks are therefore, at bottom, claims about what the classified stance tracks.

The gold-standard labels also support a stronger version of the convergent-validity estimates themselves. Because the machine-coded affinity stands in for human judgment whenever it enters a regression, I re-estimate the association between chunk-level affinity and each benchmark using design-based supervised learning. The human BWS scores are the outcome, the machine

measure is their surrogate prediction across all 66,540 chunks, and cross-fitting corrects the estimate for the surrogate's measurement error (Egami et al. 2024b).[21] The human-anchored association between affinity and UN-voting similarity is 0.36 (SE 0.09), roughly 1.6 times the 0.23 obtained when the machine measure is used naively as the outcome: measurement error in the machine labels attenuates the naive estimate, so the convergence documented above is, if anything, understated. The corresponding alliance-similarity association is positive but imprecise at the current labeled-sample size (0.42, SE 0.25, $n_{\text{labeled}} = 72$).

These divergences correspond to relationships in which the orientation a state expresses in its own strategy departs from what is behaviorally observed in foreign policy similarity. Figure 5 illustrates the NSS scores for four dyads before and after recognized inflection points in their relationships. In two of them the NSS turns warm while the observation-based measures stay cold. The United States expresses hostility toward Vietnam in its early-1990s strategies (affinity ≈ -0.57 in 1991) and then warmth once relations normalize in 1995 (affinity $\approx +0.97$). This reversal is not captured in UN voting or alliance similarity that instead both remain slightly negative throughout. The United States–Cuba relationship shows the same pattern in the period surrounding the 2014–2015 diplomatic reopening, with affinity moving from roughly -0.79 (2006) to $+0.97$ (2015) even as UN-voting similarity stays negative. In each case the NSS registers a rapprochement in close to real time that existing measures, built for symmetric and slow-moving comparisons, do not encode.

The other two dyads demonstrate cases where the NSS documents signal an antagonistic relationship that is missed with measures of observed state behavior. Alliance portfolio and UN voting similarity treat Georgia and Russia as relatively close (alliance $\kappa \approx 0.6$–0.8, UN $\kappa \approx 0.1$–0.4) by virtue of overlapping post-Soviet memberships and voting blocs. However, Georgia's

[21] The benchmark covariates in this re-estimation are the chance-corrected κ agreement indices for valued UN votes and binary alliance portfolios (Häge 2011), close analogues of the alliance and UN-voting similarity measures used elsewhere in the paper. Of the 100 gold-standard chunks, 77 merge to dyad-year benchmark covariates; the remainder lack the benchmark similarity covariates for their dyad-year — the underlying foreign-policy-similarity series does not cover every state and year in the gold sample.

rhetorical signals about its affinity toward Russia swing from mildly positive in 2005 (affinity $\approx +0.28$) to incredibly hostile after the 2008 Russo-Georgian war (affinity ≈ -0.95 in 2010). In both cases, Russia remains maximally salient. Estonia exhibits the same divergence, with its expressed affinity toward Russia falling to roughly -0.98 by 2011 despite existing measures holding steady positive values. That the measure recovers these directed, time-varying shifts suggests that its divergence from the s-scores reflects additional relational content that the symmetric and slow-moving behavior-based measures are not built to carry.

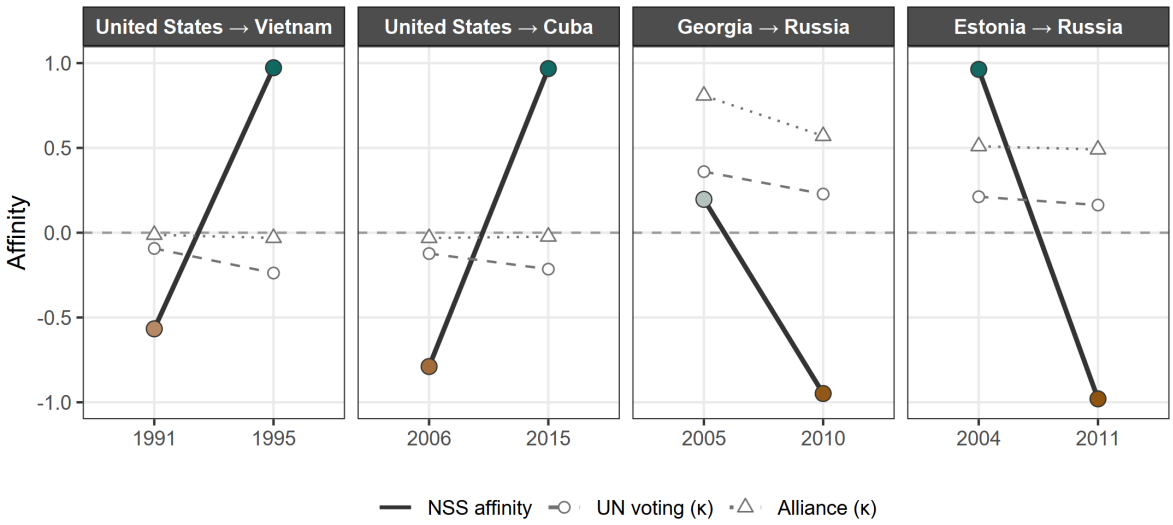


Figure 5: Dyads where NSS-expressed affinity diverges from alliance and UN-voting similarity, before and after a documented inflection in each relationship. The kappa-corrected aggregates (gray) barely move while NSS affinity reverses sign.

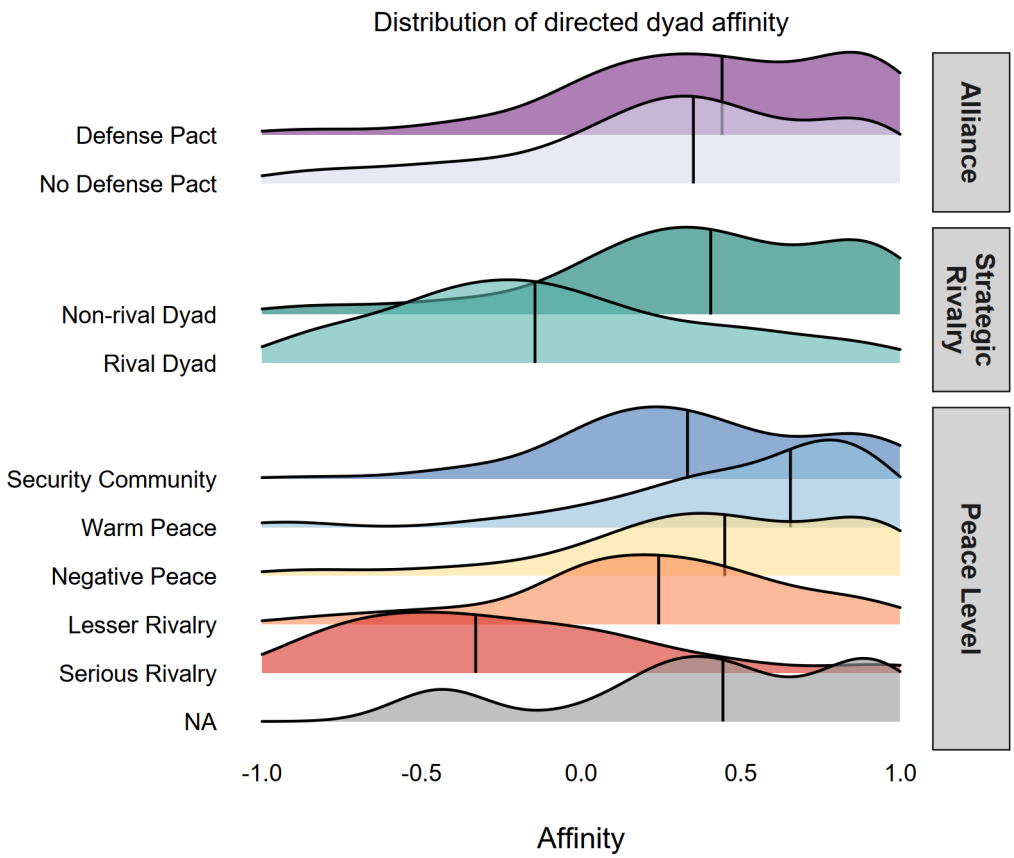


Figure 6: Distribution of NSS affinity across levels of the existing affinity measures. Vertical lines are medians. Dyads without significant ongoing interactions are coded NA in the peace-level data.

I also calculate the distribution of the NSS affinity index across three categorical measures of alignment: formal defense pact membership, strategic rivalries, and peace levels (Leeds et al. 2002; Thompson, Sakuwa, and Suhas 2021; Diehl, Goertz, and Gallegos 2021). The findings in Figure 6 show dyads in an alliance have, on average, a higher NSS affinity than dyads without a defense pact, and that dyads composed of strategic rivals have a lower NSS affinity than dyads that are not strategic rivals. The five ordered categories in the peace level data show a

clear and consistent positive correlation between peace levels and NSS affinity *until* the highest peace level – security community – which has a median NSS affinity higher than negative peace but lower than lesser rivalry. Many of the security community dyads are composed of states with shared EU membership that otherwise have little active or relevant security cooperation with one another, and may consequently have less intensely positive expressions toward each other in their NSS documents (Michalski and Norman 2016). Alternatively, deeply integrated security communities with highly institutionalized security cooperation may be largely taken for granted, lessening the perceived need to signal positive affinity and reserving more positive framings for relationships where deeper cooperation is being more actively cultivated (Gannon and Kent 2021).

6 Part III: Application

I model the association between affinity and salience and two well-regarded indicators of higher intensity interstate engagement: the initiation of militarized interstate disputes (MID) and the frequency of diplomatic visits. Using two different dependent variables that capture distinct types of state interaction – one combative and the other cooperative – produces a complementary, two-sided test of whether the relational logic expressed in NSS documents anticipates future behavior. Like the previous sections, I use the directed dyad-year as the unit of analysis, lagging the predictors by a year. The MID onset variable is drawn from the Correlates of War (COW) dataset and is coded as a binary indicator for whether the issuing state initiated a militarized interstate dispute against the target state (Jones, Bremer, and Singer 1996; G. Palmer et al. 2015). Diplomatic visits are high-level official visits between heads of state or government and are measured as an annual count using the Country and Organizational Leader Travel (COLT) dataset (Moyer et al. 2025).

I fit logistic regression models to predict MID onset and Poisson regression models to predict diplomatic visits. All models include polynomial splines accounting for temporal trends and dyad-clustered standard errors as well as control variables associated with each outcome in existing literature: CINC ratios of military power for MID onset (Singer 1988), whether either state is a great power for diplomatic visits (Lebovic and Saunders 2016), and capital to capital distance and shared regime type for both (Markowitz and Fariss 2018). Rather than asking whether the NSS measures out-predict these established indicators, I ask whether they add information beyond them. For each outcome I estimate a baseline model (M0) containing only existing measures – kappa-corrected alliance-portfolio and UN-voting similarity, and, for conflict onset, a politically-relevant-dyad indicator – alongside the controls, and a second model (M1) that adds the NSS salience and affinity measures. Because affinity is defined only for directed dyad-years in which the issuing state's NSS names the target, all models are estimated on a common sample of mentioned dyad-years, a sample selected on whether the issuing state chose to name the target. Because MID onset is a rare event, I include rare event-specific out-of-sample diagnostics including event-stratified cross-validation and precision-recall AUC benchmarked against the base rate (appendix) as well as replacing fixed effects with partial pooling (King and Zeng 2001a, 2001b).

The results, presented in Table 2, show that the NSS measures carry validated incremental information for both outcomes.[22] For each outcome, a dyad-clustered joint Wald test that salience and affinity are both zero is decisively rejected, meaning the two measures add predictive information beyond the existing benchmarks. The coefficient for salience is positive for both models meaning targets featured more prominently are more dispute-prone and more frequently visited. Affinity is positive and significant for diplomatic visits, and when it comes to predicting conflict the point estimate is predictably negative: targets described in warmer

[22] Coefficients for the control variables are omitted here, but provided in the appendix, to avoid misinterpretation since confounders for the controls variables are not included in the model (Keele, Stevenson, and Elwert 2020; Dworschak 2023).

terms see fewer subsequent disputes — a standard-deviation increase in expressed warmth lowers the odds of onset by roughly a quarter to a third, and that relationship holds across alternative constructions of the measure (appendix Table A7). As a single main effect, however, it averages over a relationship that depends on salience. I turn to the interaction between the two variables in the decomposition that follows (Section 6.2).

Because states issue these documents irregularly, salience and affinity are defined only for the years in which an origin actually publishes an NSS; values are not interpolated or carried forward between publications. The directed dyad-year panel therefore rests on 425 distinct origin-year NSS documents, each of which contributes the targets it names. I lag both dependent variables by one year so that each strategy is paired with the interstate outcomes that follow its publication rather than those that precede it.[23]

To put the M1 increments in substantive terms, I move each NSS measure from its 10th to its 90th percentile, holding the other NSS measure at its median, the controls at their means, and the year at its median. A shift of that size in salience moves the predicted probability of a subsequent MID onset by 2.4 percentage points, while the same shift in affinity moves it by -0.7 points. For diplomatic visits both measures matter: the expected visit count changes by a factor of about 1.55 across the salience range and 1.9 across the affinity range. Table 3 reports the predicted values underlying these comparisons.

The incremental information is not an artifact of in-sample overfitting. Table 4 compares M0 and M1 out of sample under cross-validation folds grouped by directed dyad, so that no dyad appears in both the training and test partitions (Cerqua, Letta, and Pinto 2025). For the rare MID-onset outcome the folds are additionally stratified on whether a dyad ever experiences a dispute. Adding the NSS measures improves out-of-sample performance for both outcomes: the area under the ROC curve for MID onset rises from 0.843 to 0.878, and the mean absolute

[23] Given that these documents are issued and revised over multi-year cycles, a one-year horizon is the shortest that cleanly separates the articulated posture from its potential consequences.

Table 2: Incremental validity of the NSS measures. Models 1–2 are logistic regressions of MID onset (odds ratios); models 3–4 are Poisson regressions of diplomatic visits (incidence-rate ratios). Controls and temporal splines are estimated but omitted, and appear in full in the appendix. Dyad-clustered 95% confidence intervals in parentheses; the final row reports a dyad-clustered joint Wald test that both NSS coefficients are zero in M1.

	MID onset		Diplomatic visits	
	M0	M1	M0	M1
Salience		50.323***		2.232***
		(13.380, 189.262)		(1.735, 2.872)
Affinity		0.445**		1.840***
		(0.253, 0.783)		(1.597, 2.120)
Alliance similarity (κ)	1.692	1.678	1.764***	1.685***
	(0.825, 3.467)	(0.847, 3.323)	(1.416, 2.197)	(1.360, 2.086)
UN voting similarity (κ)	0.125**	0.241*	0.909	0.948
	(0.032, 0.493)	(0.071, 0.824)	(0.733, 1.129)	(0.766, 1.172)
Politically relevant dyad	103.264***	59.086***		
	(13.872, 768.702)	(7.867, 443.804)		
Num.Obs.	6478	6478	7312	7312
Joint Wald test p		3.6e-12		7.2e-19

Dyad-clustered robust variance estimators; natural year splines (df = 4); controls estimated but omitted (full tables in the appendix).
Final row: dyad-clustered joint Wald test that salience and affinity are both zero in M1.

Table 3: Substantive magnitudes from M1. Predicted probability of MID onset and expected diplomatic-visit count with each NSS measure at its 10th versus 90th percentile, the other NSS measure at its median, controls at their means, and year at its median.

Outcome	Measure	At 10th percentile	At 90th percentile	Change
Pr(MID onset)	Salience	0.003	0.027	+2.4 pp
Pr(MID onset)	Affinity	0.013	0.005	-0.7 pp
Expected visits	Salience	0.33	0.51	x 1.55
Expected visits	Affinity	0.31	0.59	x 1.90

Table 4: Out-of-sample performance of M0 and M1. Cross-validation folds are grouped by directed dyad so that no dyad appears in both the training and test partitions (Cerqua, Letta, and Pinto 2025). MID onset, a rare event, is assessed by ROC-AUC and precision–recall AUC (PR-AUC) under event-stratified dyad folds and diplomatic visits by mean absolute error (MAE) and root mean squared error (RMSE) under dyad folds.

Outcome	M0 (existing)	M1 (+ NSS)	Improvement
MID onset (ROC-AUC; PR-AUC)	0.843; 0.106	0.878; 0.212	AUC +0.035
Diplomatic visits (MAE; RMSE)	0.645; 0.989	0.626; 0.980	MAE -0.019

error for diplomatic visits falls from 0.645 to 0.626. The gains are predictably modest and consistent in direction across the in-sample and out-of-sample tests.

Additional models provided in the appendix confirm that these results hold across alternative specifications. I include bivariate models that avoid post-treatment bias and over-adjustment (Clarke, Kenkel, and Rueda 2018; Montgomery, Nyhan, and Torres 2018; Dworschak 2023; Hünermund and Louw 2025) as well as models with country, dyad, and year fixed effects to account for unobserved spatial and temporal heterogeneity. I also test alternate dependent variable measures like MID participation and diplomatic visits from and to the issuing country. Consistent results are also found using panel-corrected standard errors (Beck and Katz 1995; Chiang and Sasaki 2023), lagged dependent variables to adjust for temporal autocorrelation, and zero-inflated Poisson models to account for the excess zeros in diplomatic visits.

6.1 Mention selection and the full population

A natural concern is that §6's models are estimated only on dyad-years in which the origin's NSS names the target. This restriction is not a discretionary choice but a property of the measure, as affinity is undefined when a state is silent about another. Silence conveys that a target is not salient (coded 0), but it carries no evaluative stance from which to infer affinity

Table 5: Full-population robustness. Each NSS dimension enters on its natural domain: affinity on the mention-defined sample, salience on the full population of published-NSS dyad-years, with non-mentioned targets coded salience 0. Models 1–2 predict MID onset (odds ratios); 3–4 predict diplomatic visits (incidence-rate ratios). Dyad-clustered 95% confidence intervals in parentheses; the differing Ns are intrinsic to the measures' domains.

	MID: Affinity	MID: Salience	Visits: Affinity	Visits: Salience
Affinity	0.375**		1.764***	
	(0.195, 0.723)		(1.541, 2.018)	
Salience		133.263***		2.228***
		(61.051, 290.889)		(1.887, 2.631)
Num.Obs.	6478	36 502	7312	39 755

(coded NA). Because the alliance and UN-voting s-scores are measures of foreign-policy *similarity*, the apt comparison among the NSS components is to affinity. On the mention-defined sample, affinity is significantly associated with both subsequent dispute onset and diplomatic visits (Table 5).

Salience, by contrast, is defined for every target in a published NSS and is 0 for the targets a state does not name as that is what silence conveys about prominence. It therefore extends to the full population of published-NSS dyad-years (n = 36,502 for the MID models, against 6,478 on the mention-defined sample). Re-estimated on that population, salience remains a statistically significant predictor of both MID onset and diplomatic visits, and for MID onset it achieves a lower AIC than either s-score. The salience and affinity components therefore each carry predictive signal that does not rely on restricting the sample to mentioned dyads.

6.2 Decomposing the relational signal

Entering the salience × affinity product as a single term imposes a strong restriction: that salience and affinity bear on outcomes only through their interaction, with no independent

effects. For cooperation this restriction is innocuous, because prominence and warmth both raise diplomatic engagement and so reinforce one another in the product. For conflict it is not: states fight targets with high salience but negative affinity, and those two pull against each other inside a single signed scalar. I therefore decompose the measure into salience, affinity, and their interaction.

Table 6 reports the decomposed models for three operationalizations of conflict — dispute onset, dispute count, and the use of force — each with peace-years and a temporal spline. Salience is a strong, positive predictor of conflict irrespective of affinity. The decomposition improves on the product-only specification in every case (the largest cluster-robust Wald p-value for the added components is 1.6e-04), confirming that the components carry information the interaction term discards.

What the interaction term obscures is visible in Figure 7, which plots the predicted probability of dispute onset across the salience–affinity plane. Predicted conflict is concentrated in the corner indicating high salience and hostile affinity and is near zero wherever salience is low, irrespective of affinity. The relational signal in NSS documents thus does predict conflict, but it does so through prominence and its interaction with hostility rather than through the symmetric product that serves cooperation so well.

These patterns are robust to the panel structure of the data. A crossed random-effects model with sender, receiver, and year intercepts absorbing the node-level dependence intrinsic to directed-dyadic data (Minhas et al. 2021) leaves salience ($p = 0.000$) and the interaction ($p = 0.078$) intact. The appendix reports further checks like measurement-error correction via SIMEX, a specification curve, fixed effects specifications and a machine-learning predictive benchmark. All of these specifications are consistent with the results here.

Table 6: Decomposed relational signal as a predictor of conflict. Each model enters salience, affinity, and their interaction, with peace-years and a year spline. Models 1 and 3 are logistic regressions (log-odds); model 2 is a Poisson regression of the dispute count (log-rate). Dyad-clustered standard errors in parentheses; controls omitted.

	MID onset	MID count	Use of force
Salience	3.904***	2.970***	3.731***
	(0.601)	(0.475)	(0.900)
Affinity	0.160	−0.338	−0.184
	(0.660)	(0.554)	(0.894)
Salience x Affinity	−1.490	−0.342	−0.853
	(0.912)	(0.692)	(1.327)
Num.Obs.	7146	8890	8890

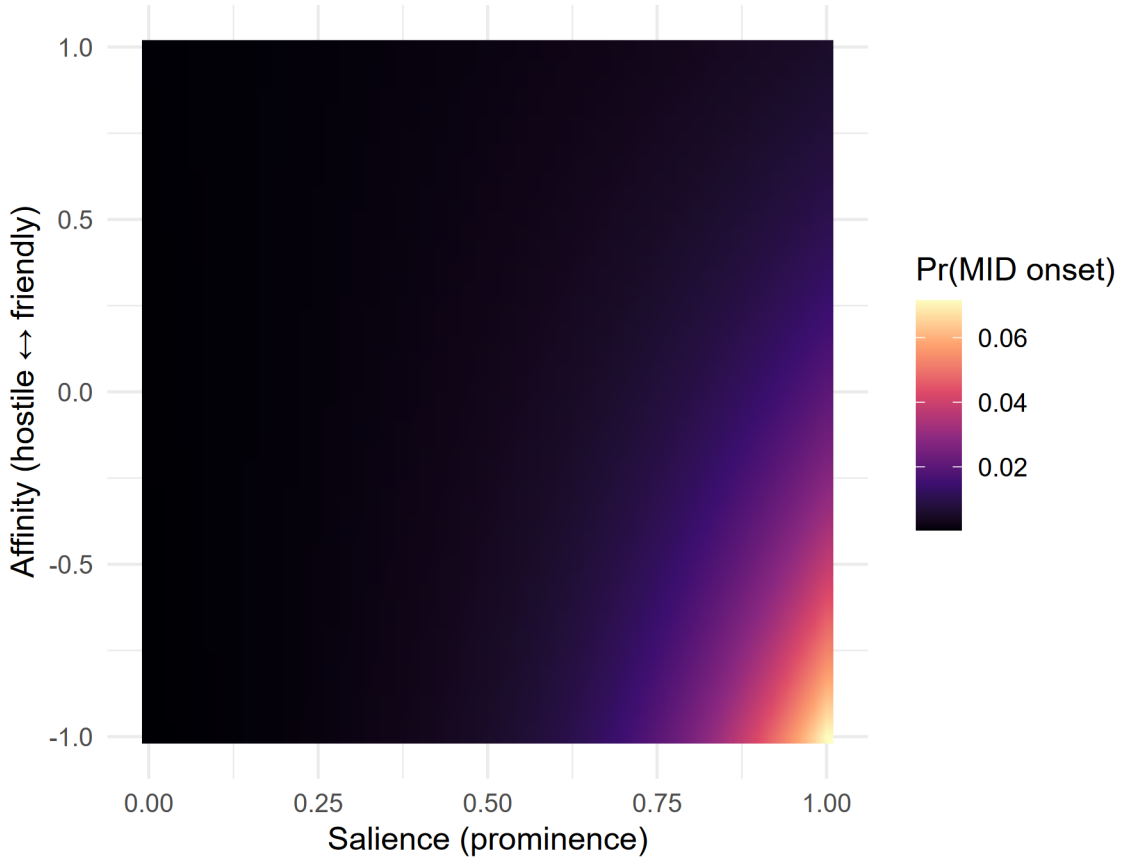


Figure 7: Predicted probability of militarized dispute onset over the salience-affinity plane. Estimated from the decomposed model with controls at their means and peace-years and year at their medians. Risk concentrates among highly salient, hostile targets and is near zero wherever salience is low.

The directedness of the measures permits one further test, reported in the appendix because the sample contains only a few dozen qualifying events. Alliance-portfolio and UN-voting similarity are symmetric — in these data the two directions of a pair never differ — so they cannot in principle address which member of a dyad *initiates* a dispute, only whether the dyad experiences one (on modeling asymmetry in dyadic relations, see Rozenas, Minhas, and Ahlquist 2019). Salience is genuinely directed, and in the appendix it predicts which side initiates: the origin state's salience toward the target distinguishes initiators where the existing measures cannot vary, a directional demonstration consistent with the asymmetry documented above.

7 Conclusion

This article makes three contributions. Methodologically, it develops a replicable pipeline that recovers directed, continuous, target-aware relational measures that identify who a text discusses, and how from any corpus where an author's orientation toward named entities matters. Empirically, it releases the most comprehensive cross-national, multi-decade collection of national security strategies in multiple human- and machine-readable formats as a public resource, together with the processing code and the derived measures. Substantively, it shows that the salience and affinity states articulate in these documents carry validated information about subsequent interstate behavior that scholar-imposed proxies do not, and that the two dimensions matter asymmetrically: cooperation is predicted by both prominence and warmth, while conflict is predicted principally by prominence, with expressed hostility contributing a smaller association.

The methodological contribution joins a live conversation. The utility and consequences of LLMs for social science questions and methods are, and will likely remain, the subject of

ongoing discussion (Bisbee et al. 2024; Laurer et al. 2024; Y. Wang 2024; Caballero and Jenkins 2025; Leo et al. 2025; Mens and Gallego 2025; Ornstein, Blasingame, and Truscott 2025; Thapa et al. 2025). My approach expands upon recent applications in political science of similar models for classification (Burnham et al. 2024), dictionary creation (Häffner et al. 2023), event extraction (Arslan, Munawar, and Cruz 2024), named entity recognition (Balluff, Boomgaarden, and Waldherr 2024), and stance detection (Burnham 2024a).

For researchers adopting these data, the analyses here support one clear recommendation: use salience and affinity as separate measures. The two dimensions have distinct external referents, validate against different benchmarks, and relate to cooperative and conflictual behavior in different ways; collapsing them into a single composite discards exactly the configurations — salient adversaries, peripheral friends — that make the measurement informative. Composite summaries may still be useful where a single scalar is required, but they should be a deliberate, theoretically motivated choice rather than the default. And when either measure enters a downstream causal design, the estimated-latent-representation problems identified for text-based measures apply with full force (Egami et al. 2022).

Used this way, the measures open questions that existing indicators cannot reach. Practitioners can harness this approach to monitor shifts in states' security discourse in near-real time, thereby supplementing traditional behavior-based indicators. And for scholars, this work opens up new avenues for research on the dynamics of interstate relationships. Further work could extend the analysis to incorporate internal policy documents like President's Daily Briefs (Goldfien and Joseph 2023; Carson, Min, and Nuys 2024) or legislative debates (Kim, Londregan, and Ratkovic 2018), thereby examining the divergence between public signaling and private deliberations (McManus 2016), or audience-specific public statements like diplomatic communiqués (Connelly et al. 2021), international agreements (Baturo, Dasandi, and Mikhaylov 2017), and diplomatic speeches in forums like the UN General Assembly (Finke and

Risse 2024; Mitra 2024; Turco 2024). These data also offer a valuable opportunity to explore how states justify signals of affinity, as such variation can be indicative of the perceived costs of hypocrisy (McManus and Yarhi-Milo 2017) or signals for a desire to deepen cooperation (Gannon and Kent 2021). The data may reveal whether individual heads of state influence the tone of bureaucratic documents (Lupton 2020, 2024) and whether public opinion predates or follows from official public statements about other states (Page and Shapiro 1983; Matush 2023), thereby linking elite rhetoric and public opinion to broader patterns of international behavior.

Beyond international relations, this research contributes to wider debates about the relationship between public rhetoric and political behavior, demonstrating how computational text analysis can bridge the gap between stated intentions and observable actions across domains of political science (Wong 2026). While I theorize only salience and affinity here, NSS documents and others like them contain a vast array of other information that has been analyzed in other contexts like the use (or non-use) of emotional and moral language (Hall 2015; Post 2023; Yoon 2025) and different rhetorical devices like persuasion (Walsh 2005; Krebs and Jackson 2007) and expressions of resolve (Trager 2010; McManus 2017).

These contributions come with scope conditions. The measures cover only states that publish an English-language, whole-of-government strategy — a selected population quantified in Section 3.2 — and only the years in which they publish, so the panel is sparser than the behavioral aggregates it complements. Although the original corpus includes 130 documents written only in states' native languages, existing computational tools can only calculate salience and affinity for texts in English; these techniques are advancing quickly, and the corpus is built to accommodate improved multilingual and stance-detection methods as they emerge. The application results are predictive associations on that selected panel, not causal estimates. And the constructs are the expressed salience and affinity a state chooses to publicly communicate,

which the divergence analyses show is related to, but not interchangeable with, the alignment its behavior reveals.

Those boundaries are also where the next questions sit: these data invite substantive work on the *social signaling* properties of security documents — how what a state publicly conveys about others operates relative to costlier signals like military deployments or formal alliances — and on what leads a state to treat particular others as politically relevant beyond the usual structural factors of geography and power.

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